Report- Kinetic Monte Carlo Simulation of hBN Growth

October 10, 2026

Saurav Mittal

Centre for Development of Advanced Computing (CDAC), Pune, India

1 Introduction

Kinetic Monte Carlo (KMC) simulations are a central tool for modeling non-equilibrium processes such as crystal growth, surface diffusion, and phase transformations across time scales inaccessible to molecular dynamics. By explicitly resolving stochastic sequences of activated events, KMC enables the study of morphology evolution driven by rare but physically decisive transitions [1, 2]. In lattice-based KMC models of two-dimensional materials, growth morphology emerges from a competition between frequent low-energy diffusion or swing processes and rare high-energy bond-breaking events whose activation reshapes island edges and controls long-term structural evolution [3].

1.1 Kinetic Monte Carlo Simulation

The kinetic Monte Carlo (KMC) simulations employed in this work follow a lattice-based, event-driven framework developed for two-dimensional hexagonal boron nitride growth,

where crystal morphology emerges from the stochastic sequencing of activated atomic processes [3]. The system evolves through a discrete sequence of events corresponding to adsorption, diffusion, desorption, and bond-preserving swing motions, with the allowed event catalogue constructed dynamically from the local bonding environment at each simulation step.

Each event i is assigned a transition rate of Arrhenius form

$$r_i = \nu_0 \exp\left(-\frac{E_i}{k_{\mathrm{B}}T}\right), \tag{1}$$

where ν_0 is the attempt frequency, E_i is the activation energy barrier associated with the event, k_{B} is the Boltzmann constant, and T is the substrate temperature. Following Ref. [3], the attempt frequency is fixed at $\nu_0 = 10^{12}\,\mathrm{s}^{-1}$, while the energy barriers are treated as configuration- dependent constants determined from first-principles nudged elastic band calculations.

At each KMC iteration, a single event is selected by sampling a uniform random number $\xi_1 \in (0, 1)$ and identifying the smallest index q such that

$$\sum_{i=1}^{q} r_i \geq \xi_1 \sum_j r_j. \tag{2}$$

The physical time increment associated with this event is then computed as

$$\Delta t = -\frac{\ln \xi_2}{\sum_j r_j}, \tag{3}$$

where $\xi_2 \in (0, 1)$ is an independent random number.

2 Methods and Analysis

To simulate the growth of hexagonal boron nitride (hBN), we employ a kinetic Monte Carlo (KMC) model defined on a 32×64 hexagonal lattice. The event catalogue includes deposition, diffusion, desorption, and low-energy swing events. Swing events correspond

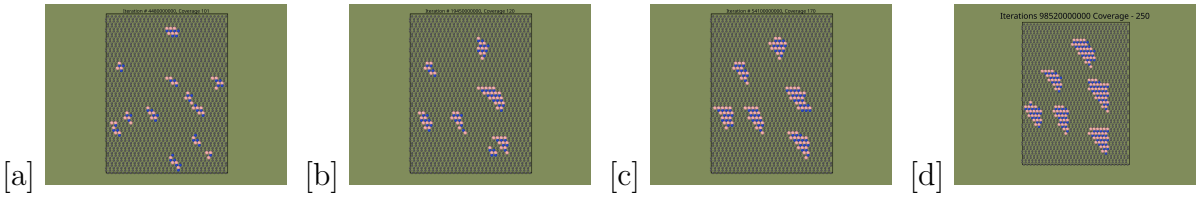


Figure 1: Representative snapshots of hBN crystal growth obtained using pseudo-random numbers at successive stages of the simulation: (a) initial seeding, (b–d) island formation and subsequent growth.

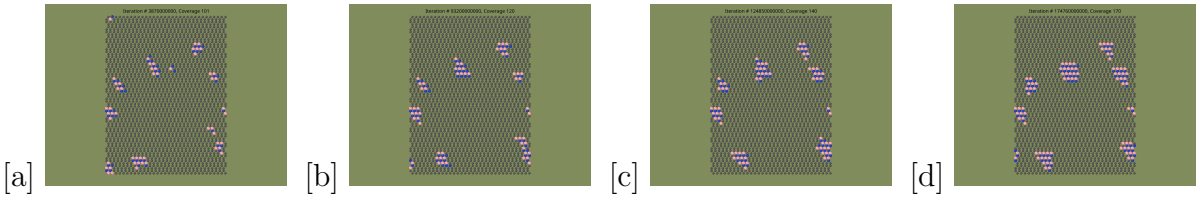


Figure 2: Representative snapshots of hBN crystal growth obtained using Quantum numbers at successive stages of the simulation: (a) initial seeding, (b–d) island formation and subsequent growth.

to local, non–bond-breaking rearrangements and therefore occur with significantly higher rates compared to bond-breaking processes. A fixed deposition flux is imposed after an initial seeding stage of 100 atoms. This procedure allows the formation of small clusters (see Fig. 1(a), Fig. 2(a)) and the system evolves according to the KMC algorithm. Subsequently, some of these small clusters merge depending on their spatial separation and later crystal islands emerge that become spatially fixed and grow along their edges (Fig. 1(b-d), Fig. 2(b-d)). While swing events dominate the event catalogue, it is the rare bond-breaking events that shape the islands. The simulations consistently exhibit two distinct growth regimes, separated by event statistics and morphology.

2.1 Crystal Growth

Initially, the simulations are dominated by swing events that do not involve bond breaking. These events occur at very high rates, leading to a large number of KMC steps with small time increments. As a result, the physical growth appears slow despite the fixed deposition flux, since many rearrangement events occur between successive deposition events. As illustrated in Fig. 3, the number of possible swing events is initially very high and the growth is slow. Rare bond-breaking events gradually modify the cluster shapes,

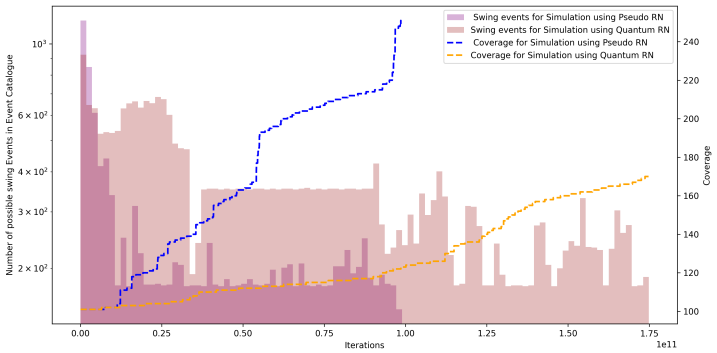


Figure 3: Evolution of swing events and coverage during KMC growth.

leading to a decrease in swing events and the emergence of distinct island morphologies for the hBN system, and the simulation shifts into the growth regime.

Once the island structure is established, the number of available swing events in the event catalogue decreases substantially. Growth then proceeds predominantly via edge processes localized at specific sites along the island boundaries. In this regime, the KMC dynamics are characterized by fewer rearrangement events between deposition events, leading to an effectively faster growth rate and stable island expansion.

References

[1] D T Gillespie *J. Phys. Chem.* **81**, 2340 (1977)

[2] A F Voter *Radiation Effects in Solids* (Springer) p 1 (2007)

[3] R Zhong MSc Thesis (Cornell University, USA) (2019)

[4] P G Bolhuis, D Chandler, C Dellago and P L Geissler *Annu. Rev. Phys. Chem.* **53**, 291 (2002)

[5] D Frenkel and B Smit *Understanding Molecular Simulation* (Academic Press) (2002)

[6] P L'Ecuyer *Handbook of Simulation* (Wiley) (1998)

[7] M Matsumoto and T Nishimura *ACM Trans. Model. Comput. Simul.* **8**, 3 (1998)

[8] A Lebedev, A Moslein, O I Yaman, D Rajan and P Intallura arXiv:2409.11539 (2024)